Knowledge Discovery Final Project

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June 27, 2026

1 Corporate Fuel Consumption: Introduction of the Dataset

This project focuses on discovering hidden conceptual knowledge within a corporate fleet's refueling operations. The dataset utilized originates from an internal company database that tracks missions and logistics in Romania. The primary objective is to analyze the `AlimentariCombustibil` dataset to discover implicit behavioral patterns among drivers and structural variations across fuel suppliers.

The original database comprised multiple tables tracking vehicles and locations, but the central focus of this research is a refined, multi-valued transactional dataset containing 2,587 unique refueling entries recorded between April 2025 and May 2026. Within the company, this table is used primarily to settle bills and deduct expenses.

1.1 Dataset Structure

While originally comprising 20 columns, the dataset was refined to include only the most applicable fields for the research. The cleaned schema of the `AlimentariCombustibil` dataset contains 8 relevant features that capture the technical and financial parameters of each refueling event. The complete column configuration is summarized below:

#	Column	Count	Dtype	Description
0	IDAlimentare	2587	int64	Unique identifier for each fuel transaction.
1	Furnizor	2587	category	The fuel supplier (e.g., OMV, Petrom, MOL).
2	Data	2587	datetime64	The exact date and timestamp of the transaction.
3	NumarInmatriculare	2587	category	The vehicle registration plate number.
4	Produs	2587	category	Specific diesel product variant selected.
5	Cantitate	2587	float64	The exact quantity of fuel filled in liters.
6	PretUnitar	2587	float64	The unit price per liter in RON (excluding VAT).
7	Valoare	2587	float64	The total monetary cost of the transaction.

Table 1: Structure and metadata of the refined `AlimentariCombustibil` dataset.

1.2 Data Cleaning and Integration Constraints

The preprocessing stage was particularly rigid due to human errors in manual records. Many registration plates contained inconsistent formats, whitespaces, or misplaced hyphens that required structural normalization.

The most significant constraint during data cleaning was that more than 50% of the initial corporate logs had to be excluded. This happened because many fleet fuel cards were assigned to general "Rezerva" (Reserve) allocations rather than specific, identifiable license plates. Since these entries cannot be linked to a single traceable vehicle profile, calculating clear individual consumption behaviors or establishing logical tracking concepts is impossible for these rows. For data processing efficiency, text fields like `Furnizor`, `NumarInmatriculare`, and `Produs` were transformed into categorical types.

To isolate the most reliable entries for advanced analytical scaling, the transactions were cross-referenced with the company's vehicle asset registry. An inner join based on registration plates identified that 23 cars from the transactional logs did not exist in the official vehicle table from the database, leading to their removal. Additionally, 15 vehicles listed in the transactions showed zero recorded fuel expenses, suggesting they represent personal cars or untracked assets.

Ultimately, filtering the data to retain only vehicles possessing internal GPS hardware resulted in a set of 55 active vehicles. Restricting the context to these 55 fully integrated assets ensures data consistency and provides a clean, validated baseline for all subsequent analytical steps and modeling stages within this research.

2 Extraction of the First Layer of Knowledge

In order to analyze the data, there are two main categories we can split the information into. The first one is more focused on the suppliers and products part, which shows the distribution of the transactions based on driver preferences regarding the types of fuel and the brands available in the network.

The second part moves to vehicle clustering, which discovers patterns and aims to categorize each vehicle into distinct clusters. Using data mining techniques, this process groups the 55 tracked vehicles based on their behavioral attributes. The clustering algorithm evaluates each vehicle using a set of operational parameters: the refuel count, total liters or money spent, average quantity filled per stop, the number of unique suppliers visited, the loyalty percentage, and the share of premium fuel selected.

2.1 Exploratory Data Analysis

This subsection provides a structured breakdown of the visualizations generated from the integrated dataset. Each analysis describes the purpose of the plot and the specific strategic insight it reveals regarding fleet management.

Proportion of Transactions by Fuel Supplier

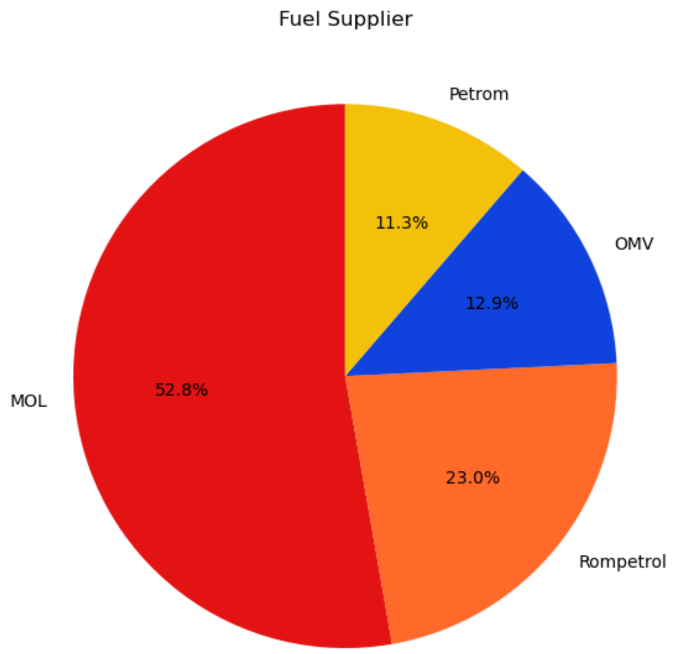


Figure 1: Fuel Supplier Piechart Distribution

A distribution analysis (Figure 1) showing which brands (OMV, Petrom, MOL, Rompetrol) capture the majority of the fleet's volume. As indicated by the numerical proportions, the dominant fuel supplier is MOL, which accounts for 52.8% of all transactions. Rompetrol follows with a significant margin at 23.0%, while OMV and Petrom represent 12.9% and 11.3% of the transaction volume, respectively.

Transaction Frequency vs. Total Value

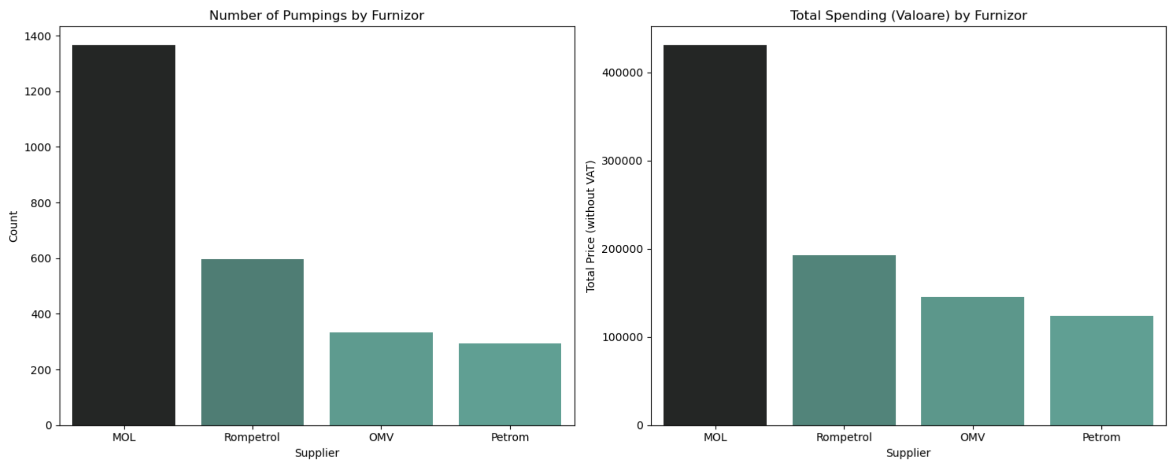


Figure 2: Two bar plots showing the transaction and the total monetary value by supplier

The dual bar chart figure (Figure 2) contrasts the number of pumpings against the total monetary value (excluding VAT) for each supplier. The left-hand plot confirms MOL as the most frequent destination with around 1,360 individual transactions, followed by Rompetrol, OMV, and Petrom. The right-hand plot displays the total accumulated cost for these transactions. As expected, the distribution remains directly proportional to the transaction count.

Distribution of Refuel Quantities

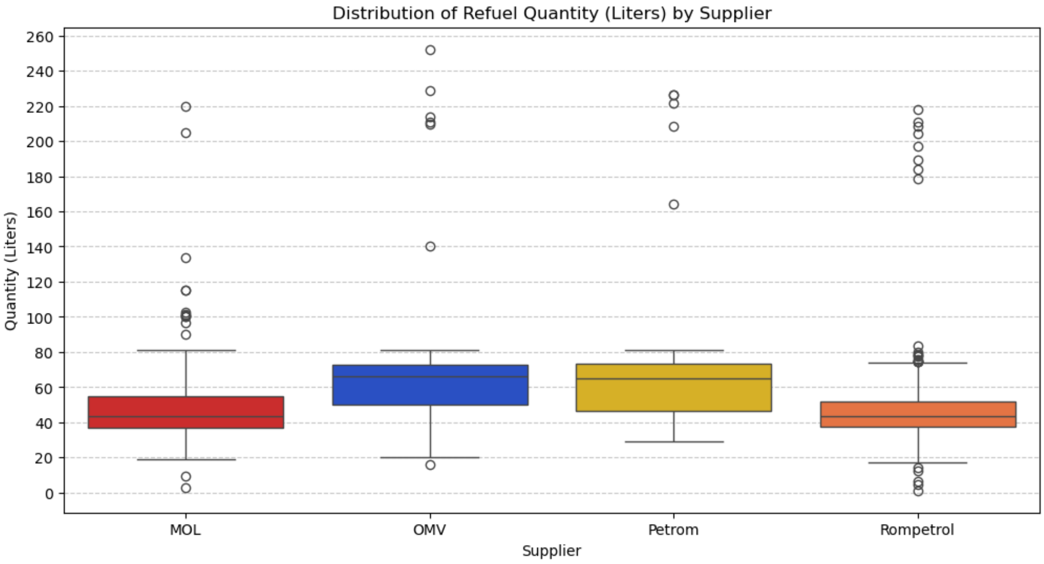


Figure 3: Refuel Quantities Boxplot Distribution

The boxplot distribution (Figure 3) provides an objective view of refueling habits across the networks. The median refill quantity for MOL and Rompetrol is consistently positioned around the **43-liter** mark. In contrast, OMV and Petrom exhibit significantly higher medians, near **67 liters**. The extended whiskers and outliers (some exceeding 200 liters) suggest a high variance in tank capacities within the fleet, specifically for vehicles refueling at OMV and Petrom stations.

Evolution of Monthly Expenses

The monthly time-series analysis (Figure 4) tracks the fluctuation of total fuel costs. The data shows distinct peaks in activity during July 2025 (70,881 RON), March 2026 (78,649 RON) and the biggest in April 2026 (85,941 RON). The visible decline in May 2026 is an artifact of the partial data recording for that month rather than a decrease in operational demand.

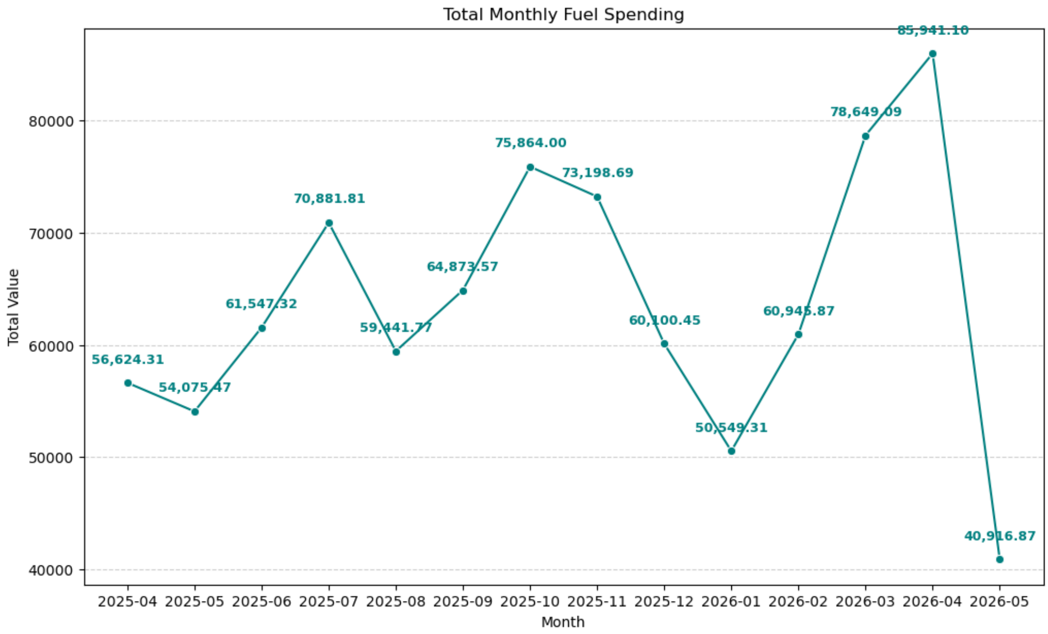


Figure 4: Monthly deducted Expenditure Lineplot

Comparative Daily Price Trends

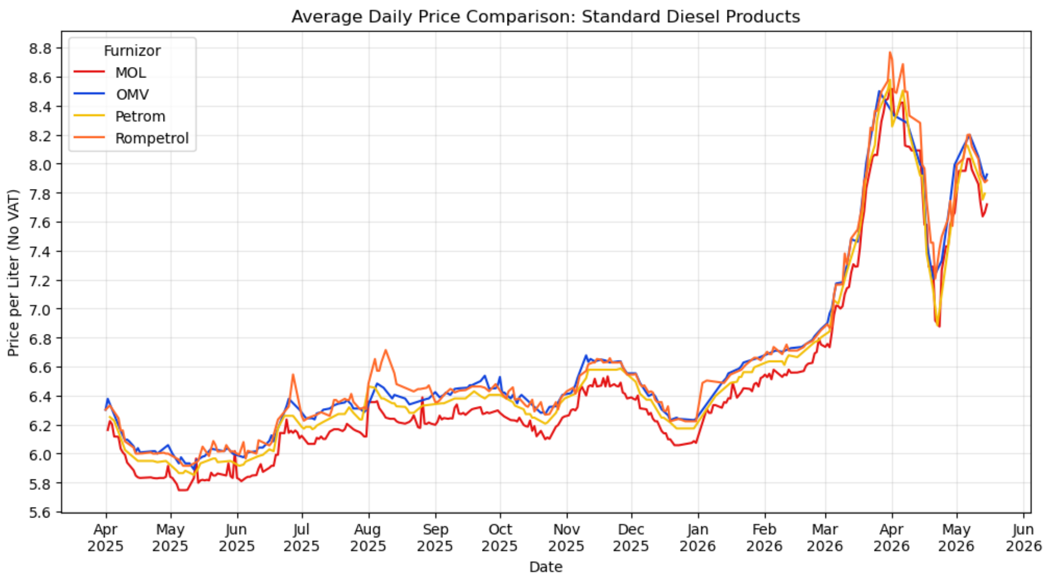


Figure 5: Standard Fuel Products Evolution Price

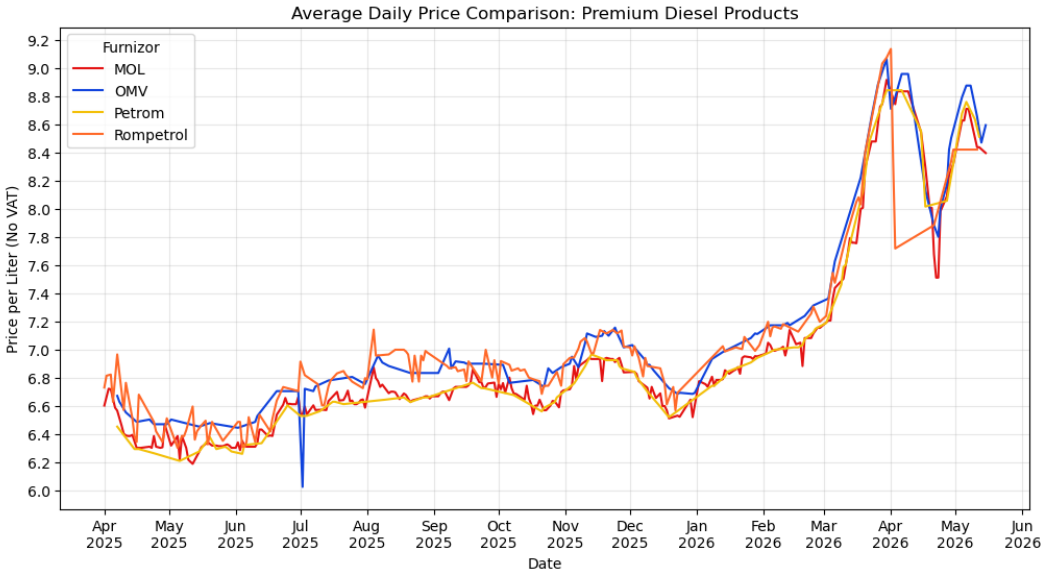


Figure 6: Premium Fuel Products Evolution Price

This line plots (Figure 5 and Figure 6) track the average daily unit price for standard and premium diesel products across all four suppliers. The visualization confirms that suppliers generally exhibit synchronous price movements in response to market changes. However, objective tracking shows that Rompetrol consistently maintains a slightly higher price floor compared to the other three suppliers, with OMV often tracking at the higher end of the standard diesel segment.

Price Dynamics of Individual Fuel Products

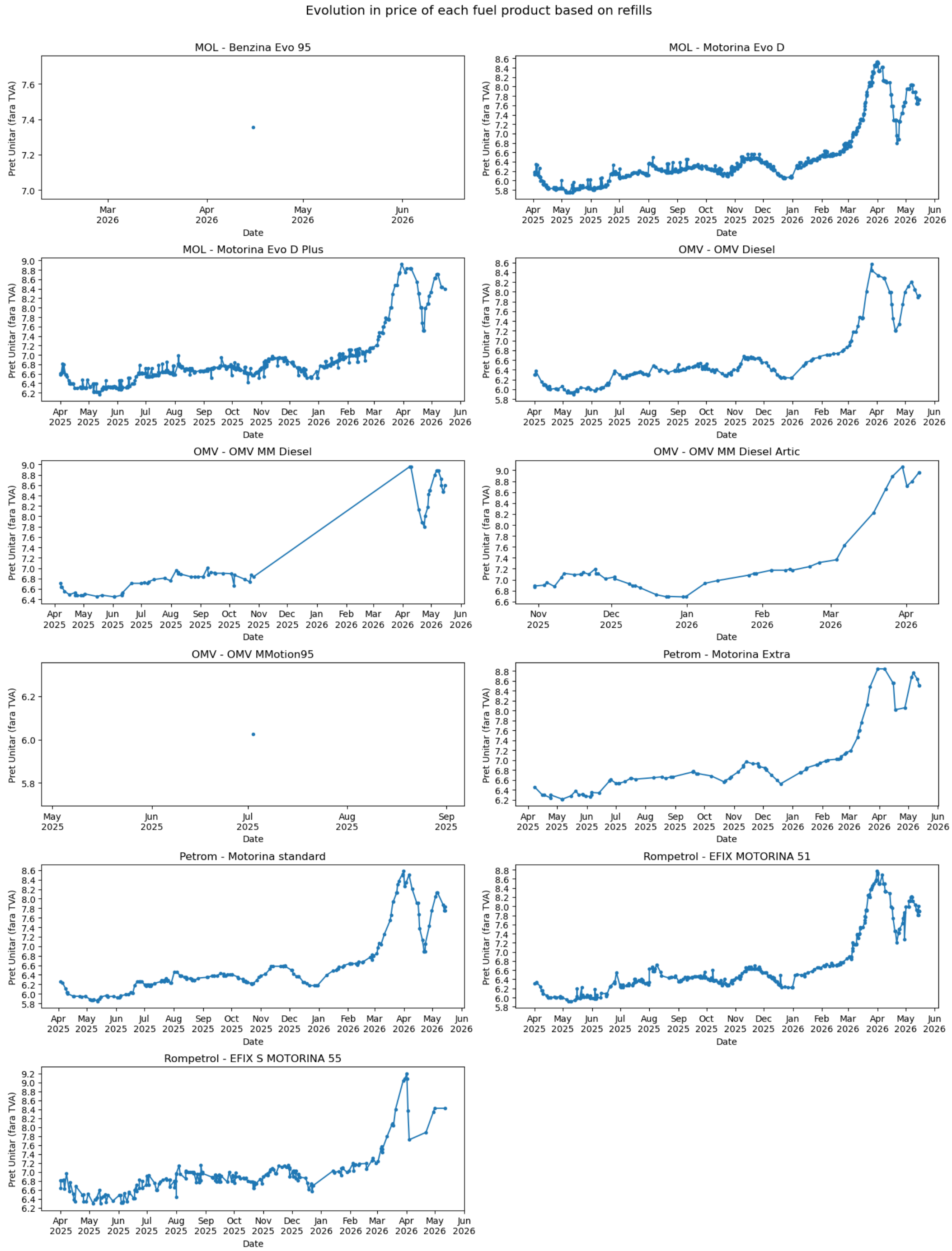


Figure 7: Individual Price Evolution of Fuel Products

The grid of subplots (Figure 7) provides a view of the price evolution for every specific product offered by the suppliers (e.g., OMV MM Diesel, Motorina Evo D etc.). These plots highlight the premium pricing tier, where products like "Efix S 55" or "Evo D Plus" fluctuate at a higher price point than standard equivalents. This objective breakdown quantifies the price gap that exists within a single supplier's portfolio.

Fleet Activity Density (Day vs. Hour)

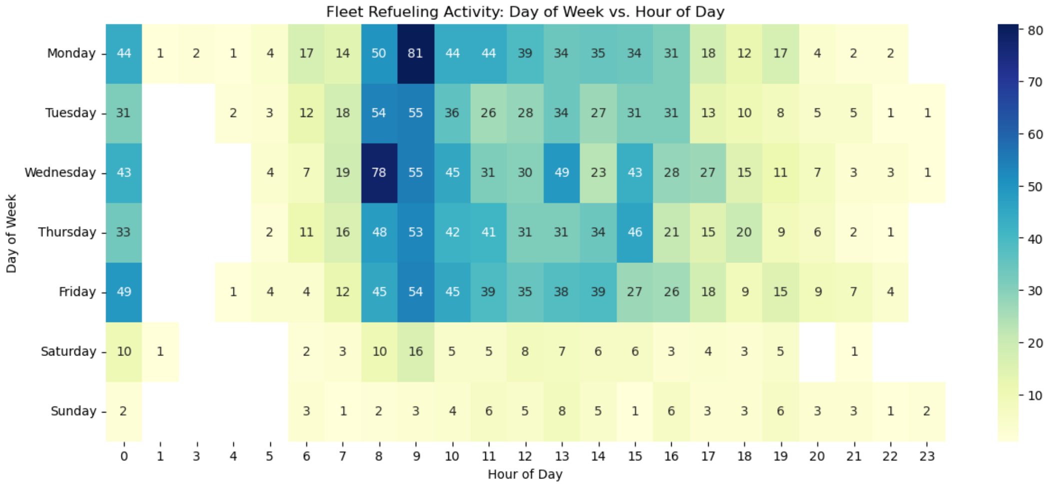


Figure 8: Heatmap: Refueling Activity by Day and Hour

The density of refueling events (Figure 8) is visualized through a heatmap correlating the day of the week with the hour of the day. The data indicates that the highest density of transactions occurs on Mondays at 09:00 AM, with activity predominantly concentrated between 08:00 and 16:00 on weekdays. Notably, refueling events recorded during nighttime hours, specifically those appearing at 12:00 AM, were identified as data entry inconsistencies.

Brand Loyalty Distribution

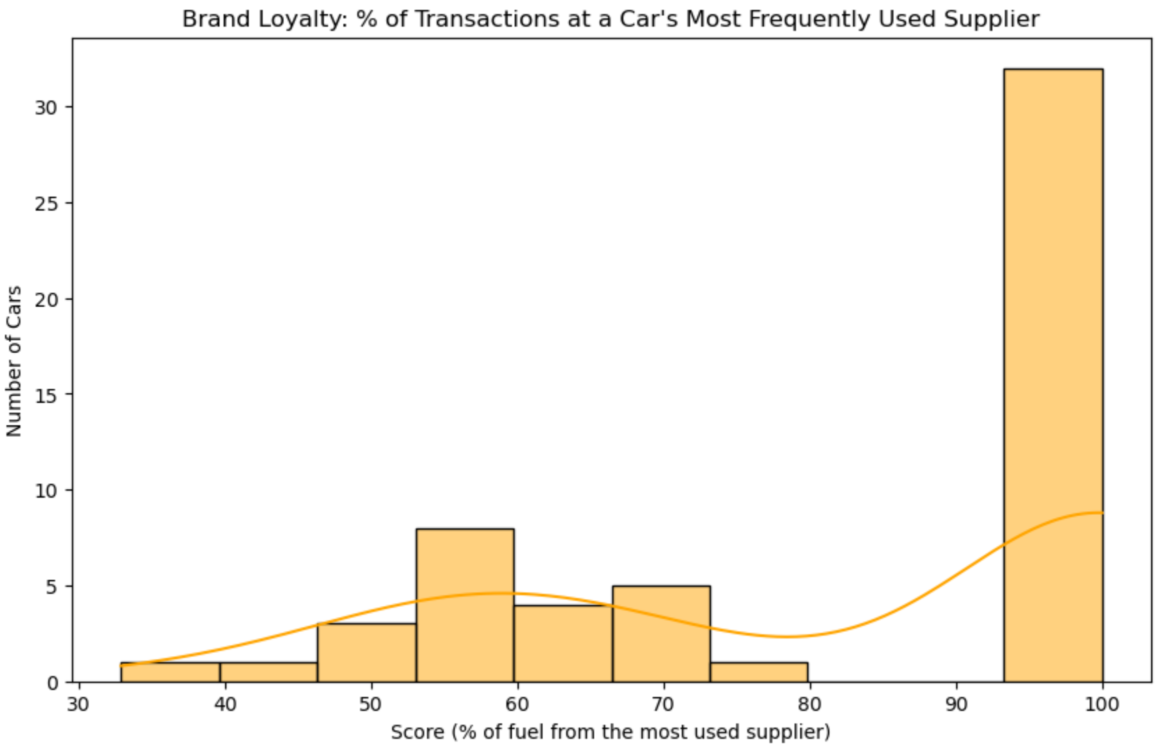


Figure 9: Bar Chart Ranking on Vehicles

The loyalty histogram calculates the percentage of a vehicle's total transactions that occur at its most-used supplier. The distribution is heavily skewed toward the right, with the median loyalty score being 100%. This indicates that a large majority of the 55 vehicles utilize only one fuel supplier exclusively, while the mean loyalty of 84.1% suggests that multi-supplier refueling is an exception rather than the rule within the fleet.

Price Sensitivity and Consumption Correlation

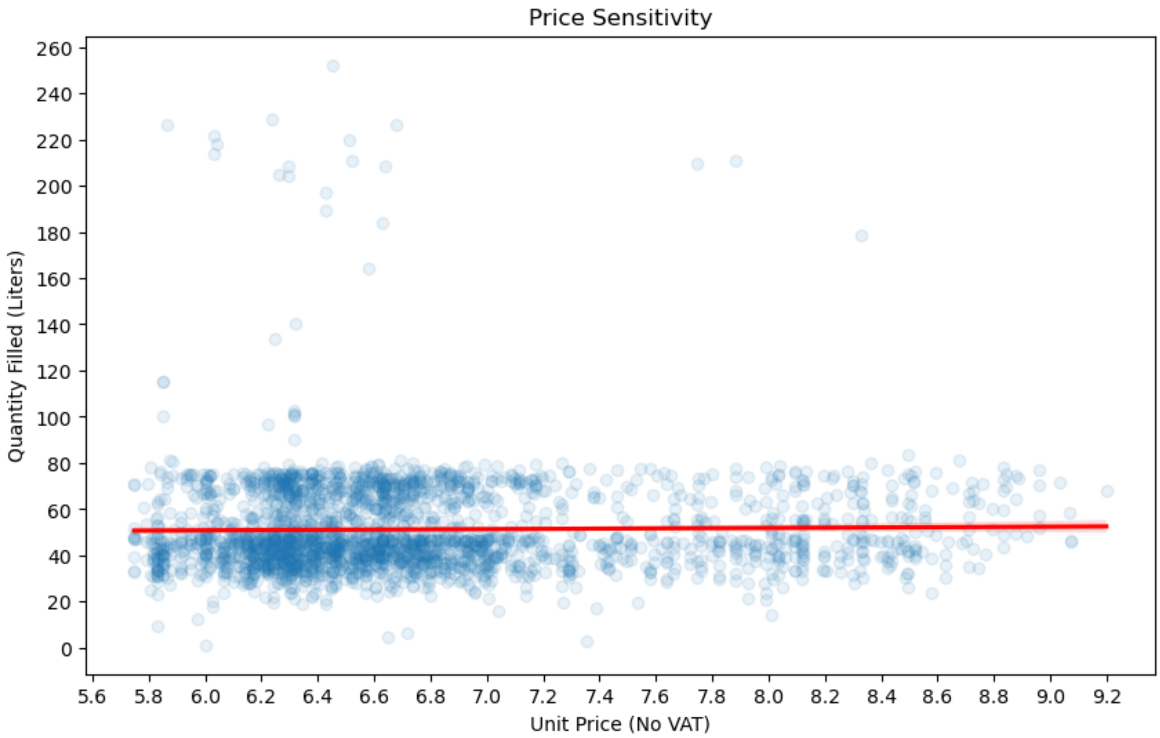


Figure 10: Bar Chart Ranking on Vehicles

The final regression plot examines whether driver behavior changes in response to fuel price fluctuations. In many corporate fleets, one might expect a flat line or zero correlation because drivers are generally price-insensitive, as the fuel is paid for by the company rather than from their own funds. While the regression plot appears dense in the left-hand region (where prices were lower), this concentration likely reflects the fact that fuel prices remained within that specific range for the majority of the recording period. Consequently, while we cannot definitively conclude that drivers are price-sensitive based on this density, the data suggests they are not entirely indifferent to price trends, even if their behavior is primarily dictated by operational necessity.

2.2 Vehicle Clustering

To transition from the transaction records to specific asset profiling, the transactional dataset was aggregated from 2,587 rows into a vehicle context containing exactly 55 rows, matching the unique registration plates tracking our fleet. For each vehicle, a set of attributes was designed: the total number of recorded refills, the cumulative sum of liters filled, the average quantity per stop, the total spending, the number of unique suppliers visited, the brand loyalty percentage (calculated as transactions at the most-used supplier divided by total refills), and the premium fuel share (the mean usage of premium lines like OMV MM Diesel, OMV MM Diesel Artic, OMV MMotion95, Motorina Extra, Motorina Evo D Plus, and EFIX S Motorina 55).

Optimal Cluster Selection and Silhouette Analysis

Before applying the clustering model, the attributes were scaled to ensure equal statistical weight. To determine the most natural grouping structure, the Silhouette score was computed to evaluate how close each vehicle is to its assigned cluster versus neighboring ones.

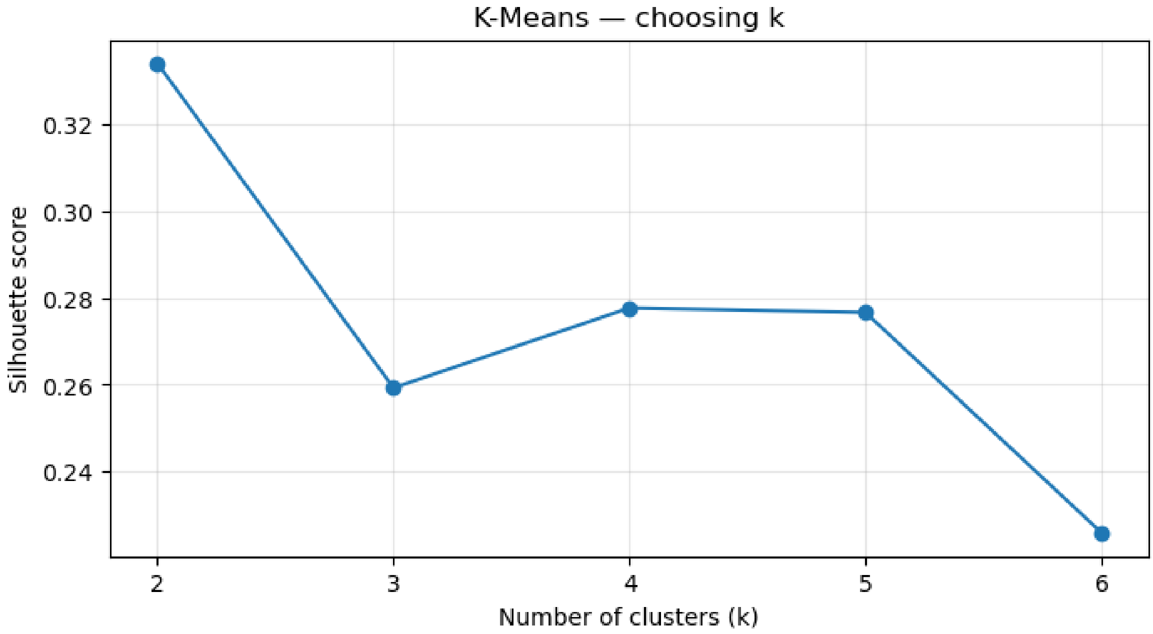


Figure 11: Silhouette Score with k=2..6

The silhouette analysis shows a visible drop in score between 2 and 3 clusters. For 4 and 5 clusters, a slight increase is observed, indicating that the model struggles to generalize cleanly or find tight, well-separated centroids within such close proximity. A much steeper decrease occurs after 6 clusters. Given that our dataset is relatively small with 55 distinct vehicle objects, we selected an optimal structure of three clusters.

Following the execution of the K-Means algorithm, the fleet settles into three distinct behavioral groups:

- **Cluster 0 — Multi-brand active ($n = 17$):** Vehicles with a high operational rate, averaging 62.8 refills and 3,850.6 liters. They show a low brand loyalty score of 56.2% and a moderate premium fuel share of 34.5%.
- **Cluster 1 — Low-volume loyal ($n = 25$):** The largest group in the fleet. These vehicles are characterized by low activity, averaging only 23.2 refills and 1,068.4 liters, but they display a high brand loyalty score of 92.7% alongside a lower premium share of 24.3%. This pattern likely indicates that these vehicles are less frequently used for long-distance business delegations, or alternatively, that their assigned fuel cards were activated much later in the recording period.
- **Cluster 2 — Premium loyal ($n = 13$):** High-intensity assets with the highest refueling frequency (72.4 refills) and 3,106.8 total liters. They exhibit an exclusive brand loyalty score of 96.8% and a dominant premium fuel share of 52.1%.

Z-Score Heatmap Interpretation

To evaluate the true variance driving these groups, a heatmap was generated by applying a standard Z-score normalization to each feature, calculated using the standard formula:

$$Z = \frac{x - \mu}{\sigma}$$

This standardizes the attributes to highlight extreme values and contrast deviations across the three clusters.

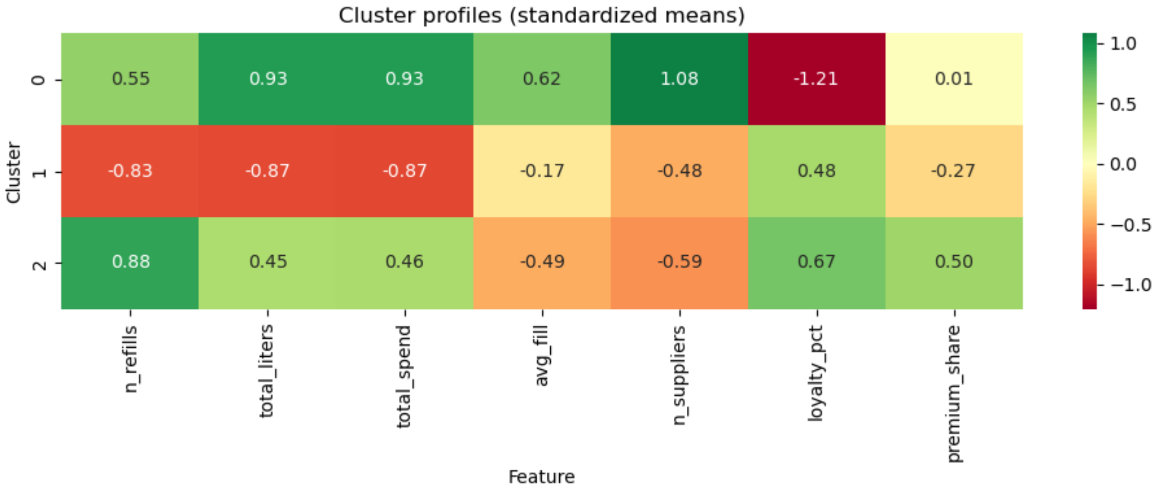


Figure 12: Z-Score Heatmap

The heatmap highlights clear operational differences between the groups. Cluster 2 has the highest positive Z-scores for both brand loyalty and premium fuel share, showing that these cars are the most dedicated to using premium products and staying with one supplier. On the other hand, Cluster 0 represents the opposite behavior, with a very low negative Z-score for loyalty. This confirms that even though they handle a large amount of fuel, these drivers have opportunistic, multi-brand refueling habits. For Cluster 1, the values are consistently below the mean for all volume, frequency, and premium attributes, but they show a clear positive Z-score for brand loyalty. This indicates that these low-use vehicles strictly stay within their assigned supplier networks.

Principal Component Analysis (PCA)

To visually verify the separation of these groups, a Principal Component Analysis (PCA) was performed to project the multi-dimensional feature space onto two dimensions. The raw variance (eigenvalues) captured by the first two components is:

$$\lambda = [3.6638, 1.6305]$$

This translates to an explained variance percentage of 51.38% for PC1 and 22.87% for PC2. Combined, these two dimensions account for more than 74% of the total variance within the vehicle context dataset.

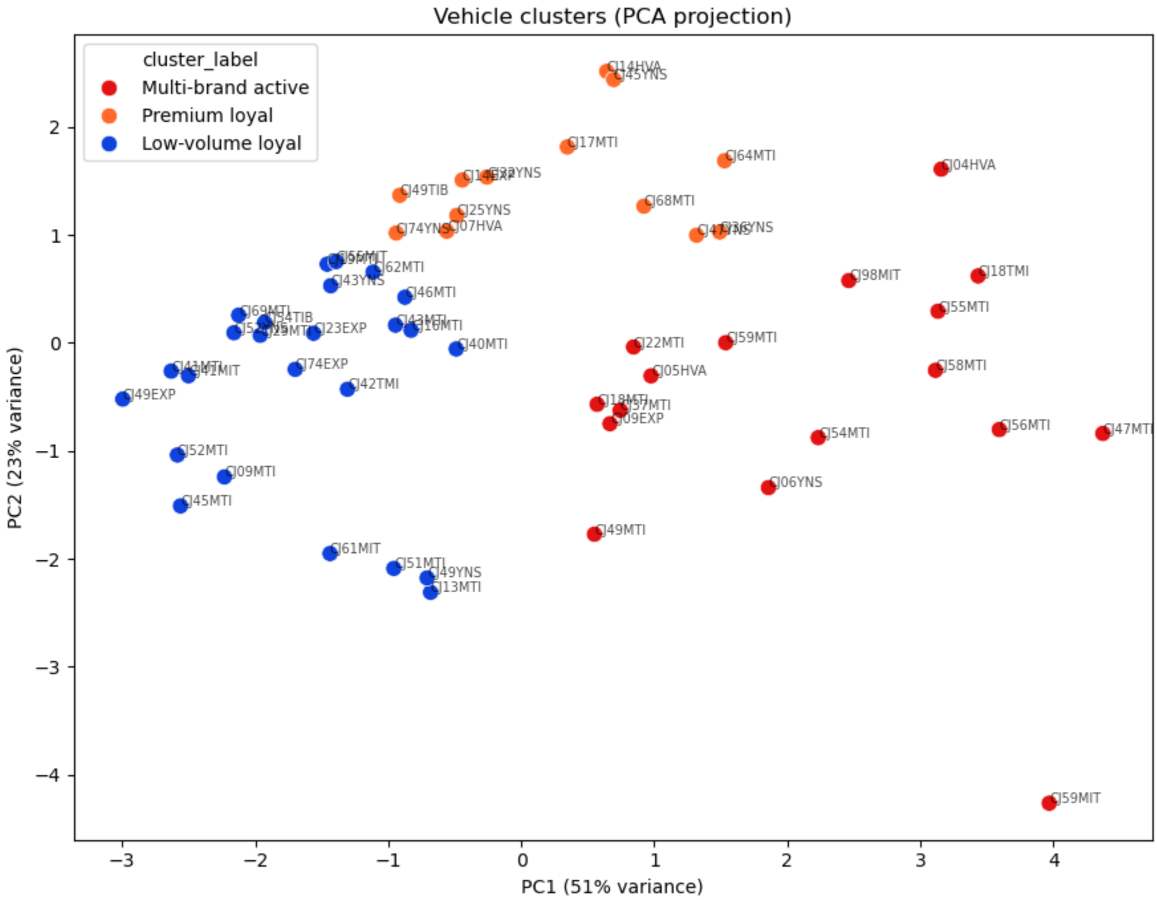


Figure 13: Clusters vizualized with PCA

The resulting PCA scatter plot shows that the three clusters are well-defined and separated in the space. There is very little overlap between the groups, which shows that the developed attributes capture clear, distinct differences in driver behavior and how the vehicles are used across the company.

3 Many Valued Context Analysis

3.1 Mapping of the Dataset's Attributes

We will split this analysis into two parts: the first one will deal with the transactions, and the second one with the vehicles.

In order to visualize the values of the attributes as a many-valued context, there should be a way to reach a consensus. For the sake of this analysis, we will remove some of the attributes because they do not contribute to the enhancement of the findings. The `Valoare` column was dropped since it is just a multiplication of the quantity of liters filled and the price of the fuel product at that time. The registration plate was also removed because, even though we could turn them into a discrete category, having 55 distinct nodes as objects would be overkill.

The datetime `Data` column was transformed into three separate parts, namely: the day type (meaning either the transaction was made during weekdays or on the weekend), the time of day (from 6:00 to 12:00 for morning, from 12:00 to 18:00 for afternoon, and the rest for evening). However, due to a processing issue in the early time of retrieving the data from the supplier records, there was an issue with the hour, minute, and second parsing, which was incorrectly taken as 12:00 AM. Because of these entries, there is a special case of "Unknown", meaning we do not know exactly in which part of the day the fueling was made. The last extracted part is the quarter of the year. Since our records start on April 1, 2025, and finish in the middle of 2026, there are 5 quarters in total.

The remaining continuous metrics were also discretized. The quantity filled was turned into a discrete form with four ranges: ≤ 25 liters, ≤ 50 liters, ≤ 80 liters, and > 80 liters. The unit price and total cost columns were segmented into four parts based on their quartiles over time. Additionally, the number of nominal values for products had to be cut down due to the potential complexity of the lattice computations, while still keeping a similar amount of information. Because of this, we mapped all the individual product types into three general categories: Standard, Premium, and Gasoline. The final structure of the transaction many-valued context is presented in Table 2.

Attribute	**Type**	**Code / Values**	**Meaning**
Id	object key	IDAlimentare	Refuel transaction id
Supplier	nominal	1=MOL, 2=OMV, 3=Petrom, 4=Rompetrol	Supplier on this refuel
ProductType	nominal	1=Standard, 2=Premium, 3=Gasoline	Fuel grade
FillSize	ordinal	1, 2, 3, 4	1: $\leq$ 25 L, 2: 25-50L, 3: 50-80L, 4: $>$ 80L
UnitPrice	ordinal	1, 2, 3, 4	PretUnitar quartiles (RON/L)
TotalCost	ordinal	1, 2, 3, 4	Valoare quartiles (RON)
DayType	nominal	1=Weekday, 2=Weekend	Day of refuel
TimeSlot	nominal	1=Morning, 2=Afternoon, 3=Evening, 4=Unknown	Hour of refuel
Quarter	nominal	1=2025Q2, 2=2025Q3, ..., 5=2026Q2	Calendar quarter

Table 2: Many-valued context attributes for fuel transactions.

For the vehicle clustering many-valued context, some of the attributes were kept from the previous analysis of transactions. The difference now is that for each car, we take the dominant value, meaning the value that was used the most often (for example, the preferred Product Tier, Supplier, DayType, TimeSlot, Quarter). Two more columns were added to the context, which are the cluster that the vehicle takes part in and the supplier loyalty ranking. The structure of this aggregated vehicle context is detailed in Table 3.

Attribute	**Type**	**Values**	**Meaning**
Id	object key	IDMasina	Vehicle identifier
Supplier	nominal	1=MOL, 2=OMV, 3=Petrom, 4=Rompetrol	Dominant supplier
ProductTier	nominal	1=Standard, 2=Premium, 3=Gasoline	Dominant fuel grade
FillSize	ordinal	1, 2, 3, 4	Mean fill (L): 1: $\leq$ 25, 2: $\leq$ 50, 3: $\leq$ 80, 4: $>$ 80
UnitPrice	ordinal	1, 2, 3, 4	Mean RON/L quartiles (55 vehicles)
TotalCost	ordinal	1, 2, 3, 4	Total spend quartiles (55 vehicles)
DayType	nominal	1=Weekday, 2=Weekend	Dominant refuel day
TimeSlot	nominal	1=Morning, 2=Afternoon, 3=Evening, 4=Unknown	Dominant refuel hour
Quarter	nominal	1=2025Q2, 2=2025Q3, ..., 5=2026Q2	Dominant activity quarter
Cluster	nominal	1, 2, 3	K-Means vehicle profile
SupplierLoyalty	ordinal	1=Opportunistic, 2=Preferential, 3=Exclusive	Top-supplier share segment

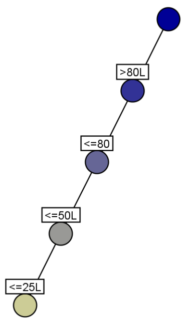
Table 3: Many-valued context attributes for aggregated vehicle profiles.

The values for the `Cluster` nominal attribute are mapped directly from our data mining results, matching the three discovered vehicle types:

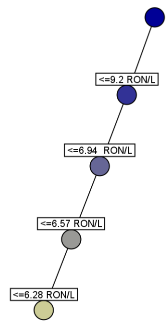
- Cluster 1 — Low-volume loyal
- Cluster 2 — Multi-brand active
- Cluster 3 — Premium loyal

3.2 Defining Conceptual Scales

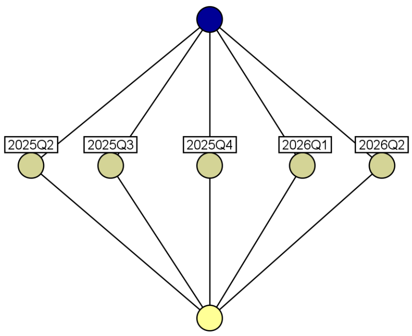
To extract structural rules from the transactional data, we configured several conceptual scales within the ToscanaJ environment. These focus primarily on the supplier-product-price area, using different types of scales to map out operational behaviors and temporal distributions.



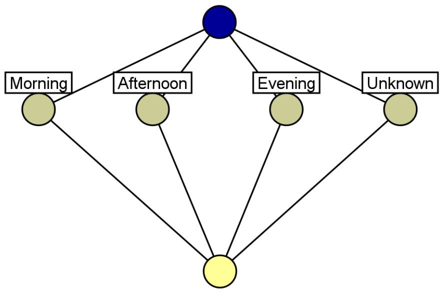
(a) Fill Size Ordinal Chain.



(b) Unit Price Ordinal Chain.



(c) Calendar Quarter Scale.

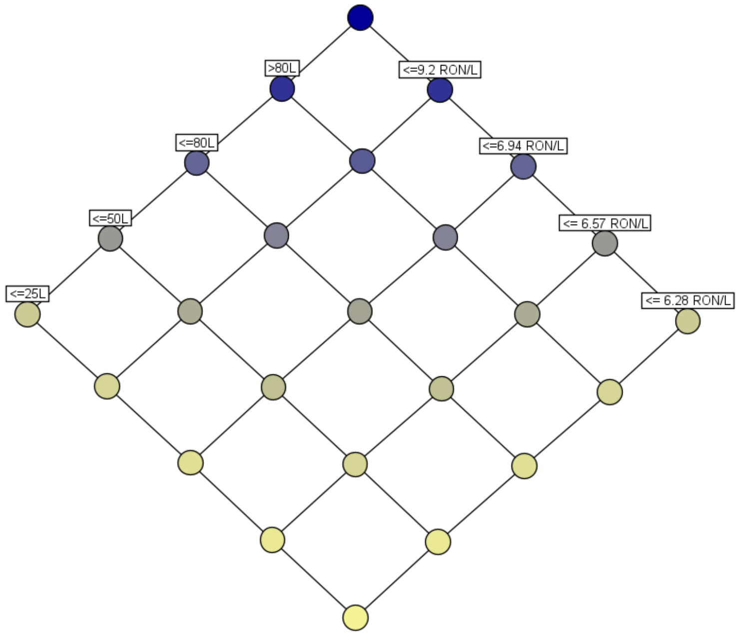


(d) Time of Day Scale.

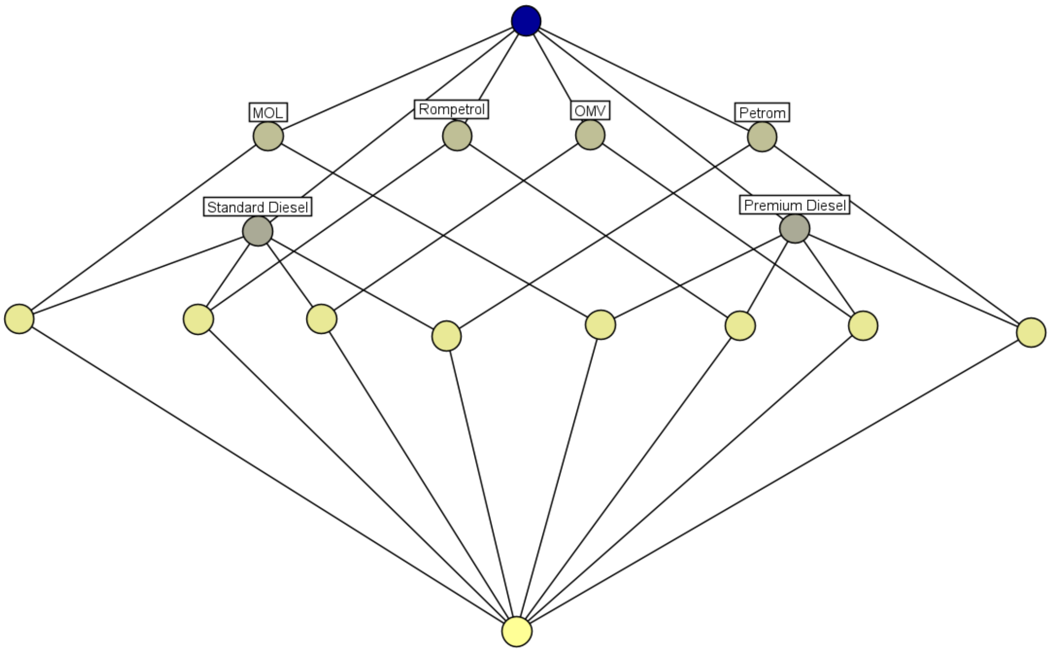
Figure 14: Nominal and Ordinal base scales for transactional parameters.

The first layer of exploration defines the fundamental metrics as single-dimension lattices, as shown in Figure 14:

- **Quantity Filled (Figure 14a):** Formed as an ordinal chain in increasing order, where each node structurally implies the ones below it. A transaction reaching the highest node ($>$ 80L) automatically satisfies the lower bounds ($\leq$ 80L, $\leq$ 50L, and $\leq$ 25L).
- **Unit Price (Figure 14b):** Another ordinal chain based on the price thresholds (6.28 RON/L, 6.57 RON/L, 6.94 RON/L, and 9.20 RON/L). This lets us track the financial escalation of transactions.
- **Temporal Scales (Figures 14c and 14d):** Configured as nominal flat contexts where attributes like the calendar quarters (2025Q2 to 2026Q2) or daily shifts (*Morning*, *Afternoon*, *Evening*, and *Unknown*) radiate independently from the top node. They act as distinct categories that do not overlap.



(a) Grid scale intersecting Volume and Price.



(b) Custom context for Supplier and Product Tier.

Figure 15: Multi-dimensional cross-cutting concept lattices.

By crossing these individual properties, we can look deeper into the structural connections within the dataset, as displayed in Figure 15:

When crossing the ordinal scales of `FillSize` and `UnitPrice`,a grid scale with 16 possible concept intersections is created (Figure 15a). Navigating this lattice allows us to see how fill sizes behave under different pricing conditions. For instance, by tracking the lines, we can verify whether large commercial fills (> 80L) still occur during peak market inflation rows (≤ 9.20 RON/L), or if transactions shrink toward the left side of the diamond (≤ 25L) when prices climb past the lower quartiles.

Figure 15b shows a custom nominal distribution context that maps our four main suppliers

(*MOL*, *OMV*, *Petrom*, *Rompetrol*) against the fuel types (*Standard Diesel* and *Premium Diesel*) (Figure 15b). The intermediate nodes show the clear intersections where drivers select a specific brand and fuel grade.

Moving on to the vehicle-level analysis, the focus shifts to evaluating how the statistical vehicle categories correspond with driver refueling loyalty. By pairing the mathematical groups from the clustering phase with behavioral consistency metrics, we can visually examine the cross-cutting relationships that define our fleet's operational alignment.

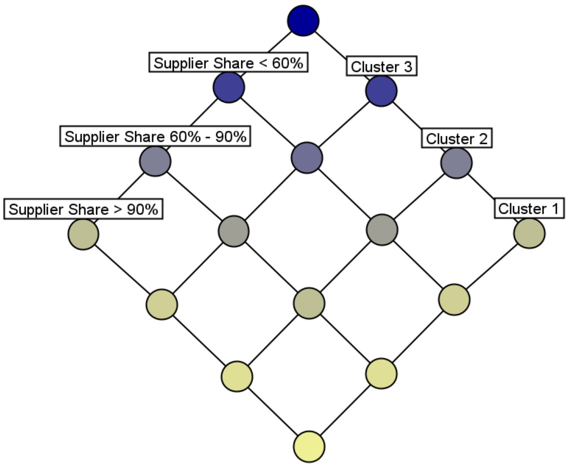


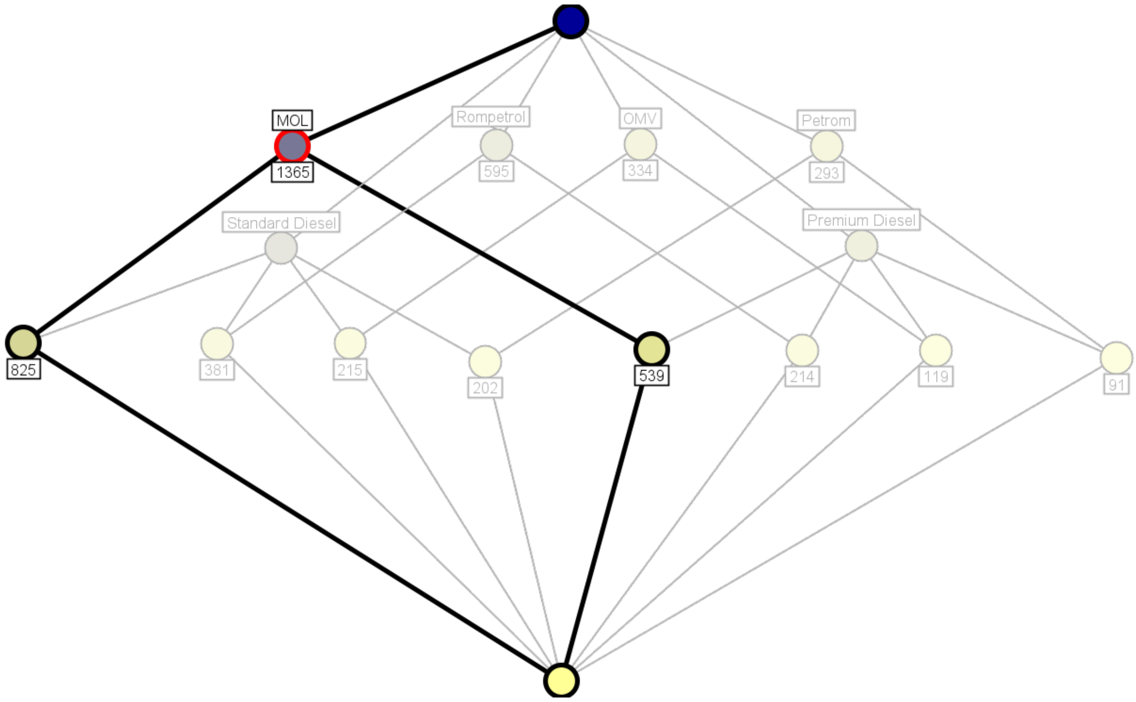
Figure 16: Grid scale intersecting Vehicle Clusters and Supplier Share.

As shown in Figure 20, this relationship is modeled using a structured grid scale that maps the three distinct vehicle clusters (*Cluster 1*, *Cluster 2*, and *Cluster 3*) against the three ordinal levels of supplier share dominance ($< 60\%$, $60\% - 90\%$, and $> 90\%$).

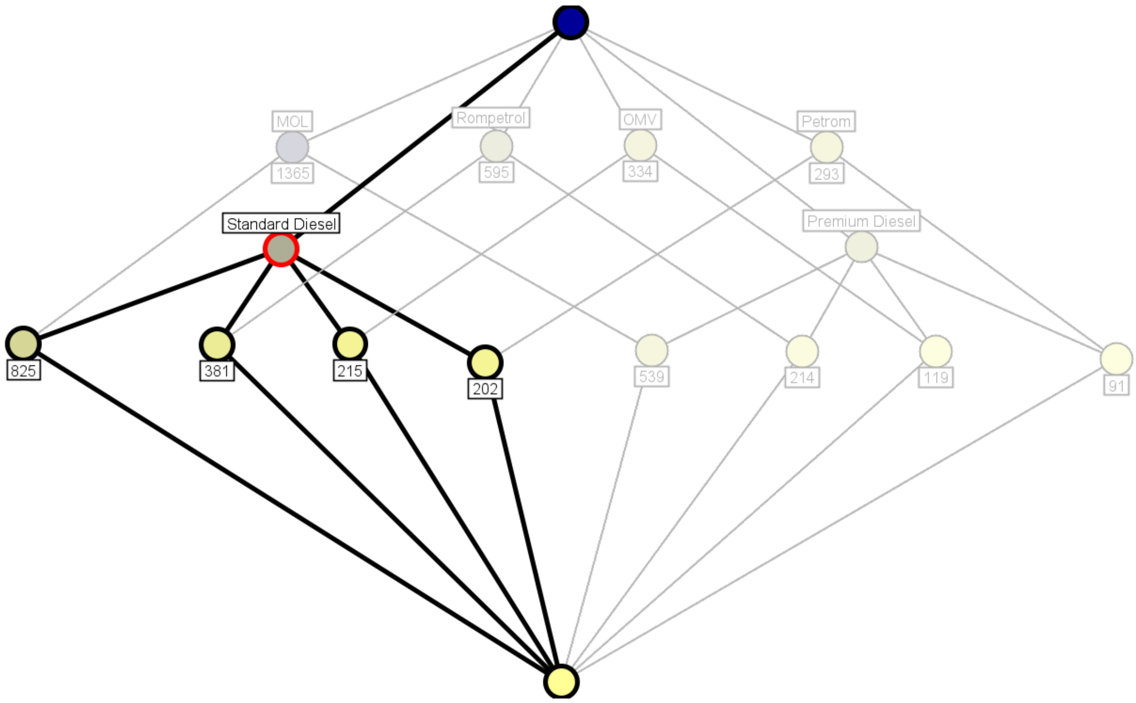
After a careful review of the initial statistical results, only a specific subset of attributes was chosen for the creation of these formal conceptual scales. Several alternative lattices were tested during the design phase but ultimately omitted from this section because they failed to reveal any new or meaningful insights. For example, pairing highly predictable parameters, such as total expenditure against fill sizes, merely repeated the obvious findings from the exploratory data analysis.

3.3 Diagram Navigation

To evaluate the core transactional relationships, we analyze the distribution of fuel choices across the main suppliers using the concept lattices. Figure 17 shows the count distribution for each supplier, split between standard and premium product tiers.



(a) Supplier and product distribution lattice.

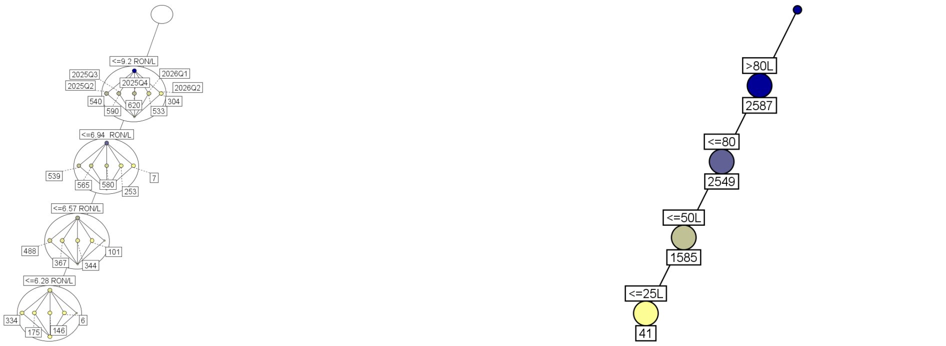


(b) Detailed node view of supplier hierarchies.

Figure 17: Concept lattices for Supplier and Product Tier transactions.

The lattice organizes the suppliers in decreasing order based on transaction volume, from the brand with the most entries down to the one with the least. Looking at the concept nodes, the number of standard diesel transactions clearly surpasses the number of premium ones across the entire network. However, the volume of premium refuelings remains significant. This indicates that a notable portion of drivers choose the premium version, likely when preparing for longer transit destinations or demanding business missions.

By navigating the main concepts, we can see that MOL is by far the most preferred supplier in the dataset. This dominant position suggests two possible operational factors: either there is a clear habit or company preference for drivers to stop at this specific brand, or the transit routes driven during company missions simply contain a much higher concentration of MOL gas stations along the Romanian road network.



(a) Nested diagram of Quarters within Unit Price. (b) Volumetric distribution of transactions.

Figure 18: Evaluation of price escalations and tank capacities.

To see how market forces impacted the fleet over time, we analyze a nested diagram that splits each price node into a nominal scale of calendar quarters. As expected, a major price explosion happened in the first quarter of 2026, driven by conflicts in the Middle East. The shift is visible: the 2026$Q2$ branch shows only 7 transactions under the ≤ 6.94 RON/L threshold, while the 2026$Q1$ branch jumps to a whopping 304 transactions under the maximum ≤ 9.20 RON/L node. This represents an increase of almost 300 transactions forced into the highest cost bracket.

In the diagram on the right (Figure 18b), we see the quantity distribution. The lattice shows a huge number of transactions between 25 and 80 liters (with 1,585 entries at ≤ 50L and 2,549 at ≤ 80L). This means drivers usually choose to fuel up the whole tank, taking into consideration the average capacity of a normal car tank.

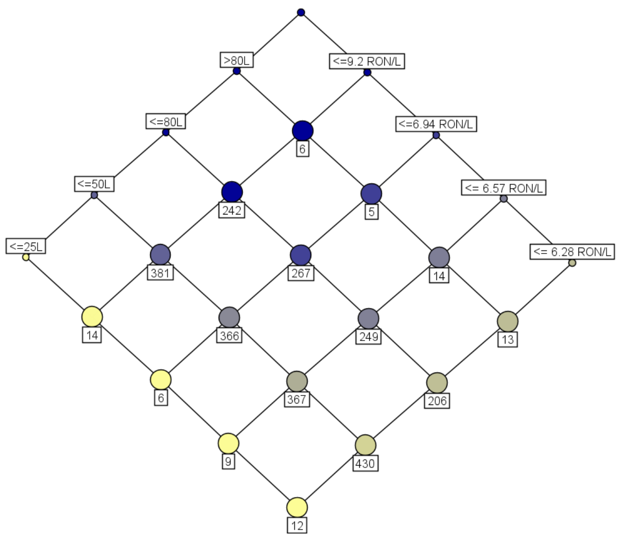


Figure 19: Concept lattice showing the intersection of Fill Size and Unit Price.

As seen in the lattice, the highest counts are concentrated along the central diagonals, specifically where the ≤ 50L and ≤ 80L nodes intersect the middle price tiers (such as the nodes with 381, 366, and 267 transactions). This distribution shows that the number of transactions per price category is almost directly proportional across the standard tank volumes. The extreme top nodes remain very small, with only 6 transactions reaching the absolute highest price ($\leq$ 9.20 RON/L) and volume (> 80L) combination, confirming that massive commercial fills during peak inflation are rare.

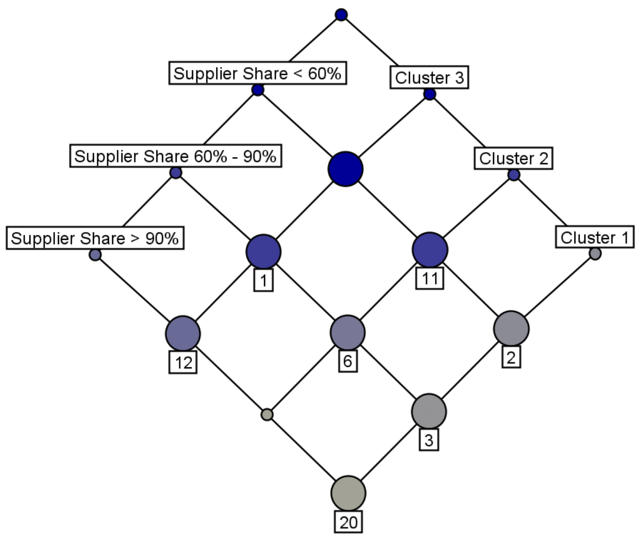


Figure 20: Concept lattice showing the intersection of Fill Size and Unit Price.

The lattice shows a clear link between how much a car is used and how loyal the driver is. Vehicles in Cluster 1 (low-volume loyal) and Cluster 3 (premium loyal) mostly sit at the bottom with high loyalty, where 20 cars have a top-supplier share of more than 90%. On the other side, the active cars in Cluster 2 (multi-brand active) stay on the left part of the grid, with 11 vehicles dropping below the 60% supplier share because they just fuel up wherever it is convenient.

4 Attribute Exploration

To test attribute exploration, we first tried using a small sample of 25 transactions from the fleet dataset so we would not surpass the limitations of the ConExp software.

However, this transactional data quickly showed major limitations and proved that it is not a good source for attribute exploration. Because these are just random transactions happening over time, the generated implications do not necessarily have a logical or real-world sense. Any rule we find depends completely on the small sample of data we happened to choose for this specific context, rather than representing true general knowledge or strict operational laws.

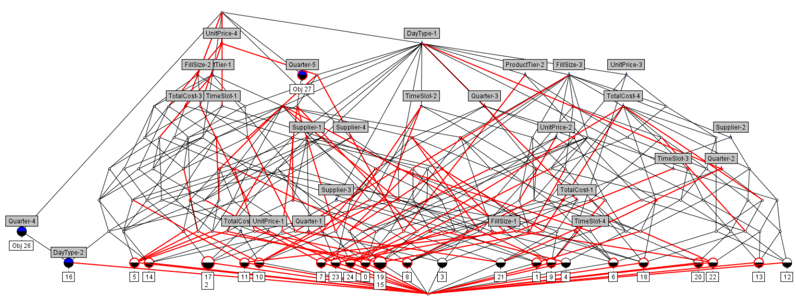


Figure 21: Concept lattice of 25 Transactions

To overcome the limitations of the transaction logs, a custom dataset focused on Machine Learning Algorithms and Architectures was built, containing 16 well-known models across 11 architectural attributes. The complete formal context is presented in Table 4.

Algorithm / Architecture	**Sup**	**Unsup**	**Self-Sup**	**Class**	**Reg**	**Clust**	**DimRed**	**Inst**	**NonLin**	**Seq**	**Tree**
SVM (RBF Kernel)	X			X	X			X	X		
Random Forest	X			X	X				X		X
K-Means Clustering		X				X					
Hierarchical Clustering		X				X		X	X		
t-SNE		X					X	X	X		
PCA		X					X				
Naive Bayes	X			X							
Logistic Regression	X			X							
K-Nearest Neighbor (KNN)	X			X	X			X	X		
DBSCAN		X				X		X	X		
Autoencoders		X	X				X		X		
Transformers			X	X	X				X	X	
Decision Trees	X			X	X				X		X
XGBoost / Gradient Boosting	X			X	X				X		X
CNNs (Convolutional)	X			X	X				X		
Neural Networks (MLPs)	X			X	X				X		

Table 4: Formal context for Machine Learning Algorithms and Architectures.

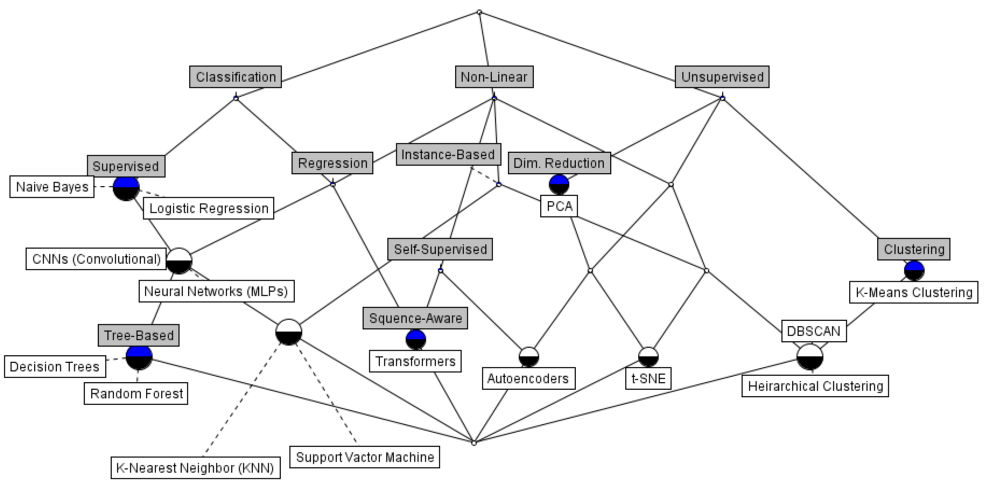


Figure 22: Concept lattice representation of the Machine Learning Algorithms data context

Using this dataset, the ConExp expert system ran a complete attribute exploration process. Instead of just listing out every single question, the interactive exploration steps and the expert decisions are organized inside Table 5.

No.	Implicational Question (Premise ⇒ Conclusion)	Conf.	Decision	Counterexamples
1	Supervised ⇒ Classification	100%	Accepted	None
2	Classification ⇒ Supervised	89%	Validated	K-Means, t-SNE, etc.
3	Classification, Non-Linear ⇒ Regression	100%	Accepted	None
4	Regression ⇒ Classification, Non-Linear	100%	Accepted	None
5	Classification, Regression, Non-Linear ⇒ Supervised	86%	Validated	Transformers
6	Instance-Based ⇒ Non-Linear	100%	Accepted	None
7	Clustering ⇒ Unsupervised	100%	Accepted	None
8	Dim. Reduction ⇒ Unsupervised	100%	Accepted	None
9	Unsupervised, Clustering, Non-Linear ⇒ Instance-Based	100%	Accepted	None
10	Self-Supervised ⇒ Non-Linear	100%	Accepted	None
11	Classification, Regression, Instance-Based, Non-Linear ⇒ Supervised	100%	Accepted	None
12	Tree-Based ⇒ Supervised, Classification, Regression, Non-Linear	100%	Accepted	None
13	Unsupervised, Self-Supervised, Non-Linear ⇒ Dim. Reduction	100%	Accepted	None
14	Self-Supervised, Classification, Regression, Non-Linear ⇒ Sequence-Aware	100%	Accepted	None
15	Sequence-Aware ⇒ Self-Supervised, Classification, Regression, Non-Linear	100%	Accepted	None
16	Supervised, Self-Supervised, Class., Reg., Non-Linear, Seq. ⇒ All Attributes	0%	Rejected	Concept Boundary
17	Supervised, Class., Reg., Instance, Non-Linear, Tree ⇒ All Attributes	0%	Rejected	Concept Boundary
18	Unsupervised, Classification ⇒ All Attributes	0%	Rejected	Concept Boundary
19	Unsupervised, Clustering, Dim. Reduction ⇒ All Attributes	0%	Rejected	Concept Boundary
20	Self-Supervised, Instance-Based, Non-Linear ⇒ All Attributes	0%	Rejected	Concept Boundary

Table 5: Exploration table of shared implications from the ConExp expert system.

A fundamental baseline rule is established by implication 1, showing that Supervised status leads to Classification with 100% confidence. Interestingly, the reverse path in implication 2 yields an 89% confidence score because exceptions exist. While most classification systems are supervised, modern self-supervised setups like Transformers handle classification without relying on traditional supervised labels.

The mathematical nature of the learning goals shows an absolute symmetry in the dataset through implications 3 and 4. The attributes of Regression and the combination of Classification with Non-Linear properties imply each other with 100% confidence. This means that every regression model chosen in the context, including neural architectures and tree ensembles, inherently processes non-linear classification data as well, emphasizing the shared architectural backend of these tasks.

For the data structures and learning paradigms, distinct design trends appear. Tree-Based models completely map out a fixed set of features, showing that they are supervised tools capable of both classification and regression within non-linear boundaries. On the unsupervised side, Clustering and Dimensionality Reduction always lead to an Unsupervised classification with a strict 100% confidence rating.

Rules 16 to 20 show the absolute boundaries of our context table. These are cases where the combination of properties yields 0% confidence because the premise combinations are physically impossible or conflicting within the data context . These combinations are automatically rejected.

5 Triadic Concept Analysis

The dataset used in this chapter focuses on the comparison of multiple benchmarks across the most relevant Large Language Models from 2025–2026 [Taqi(2026)]. To accomplish this knowledge discovery task from a total of 24 models and 31 columns, the remaining context is constituted from 6 LLM models as objects and 3 attributes covering performance benchmarks. These benchmarks are MMLU (consisting of multiple-choice questions spanning 57 different tasks), SWE-bench (specialized in real-world software engineering tasks), and HumanEval (specifically designed for Python-generated code). The condition is the accuracy, categorized into 3 nominal scales: low (L), medium (M), and high (H).

From a continuous form, the results of the benchmark for each model need to be converted into a discrete form. To ensure a distributed value mapping across the entries, different ranges are set for each benchmark:

- **MMLU:** $x \geq 90 \Rightarrow H$, $85 \leq x < 90 \Rightarrow M$, $x < 85 \Rightarrow L$
- **SWE-bench:** $x \geq 75 \Rightarrow H$, $55 \leq x < 75 \Rightarrow M$, $x < 55 \Rightarrow L$
- **HumanEval:** $x \geq 94 \Rightarrow H$, $90 \leq x < 94 \Rightarrow M$, $x < 90 \Rightarrow L$

Formally, the triadic context is defined as $K = (G, M, B, Y)$, where:

$$\begin{aligned} G &= \{\text{Claude Opus 4.5}, \text{Gemini 2.5 Pro}, \text{MiniMax-M2.5}, \text{DeepSeek R1}, \text{Mistral Large 3}, \text{Mistral Medium 3}\} \\ M &= \{\text{MMLU}, \text{SWE-bench}, \text{HumanEval}\} \\ B &= \{L, M, H\} \end{aligned}$$

The cross-cutting relations are split into three distinct slices representing each condition block within a single line.

Condition: Low (L)			
Object	**MMLU**	**SWE**	**HE**
Claude Opus 4.5			
Gemini 2.5 Pro			
MiniMax-M2.5			
DeepSeek R1		X	
Mistral Large 3		X	X
Mistral Medium 3	X	X	X

Condition: Medium (M)			
Object	**MMLU**	**SWE**	**HE**
Claude Opus 4.5			X
Gemini 2.5 Pro	X		X
MiniMax-M2.5	X		X
DeepSeek R1	X		
Mistral Large 3	X		
Mistral Medium 3			

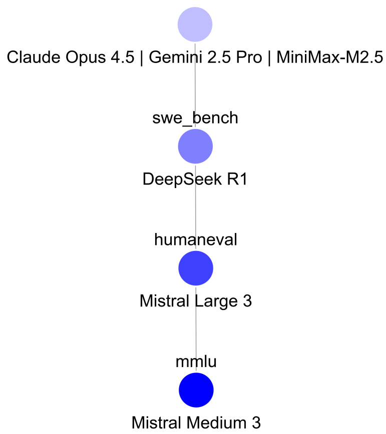
Condition: High (H)			
Object	**MMLU**	**SWE**	**HE**
Claude Opus 4.5	X	X	
Gemini 2.5 Pro		X	
MiniMax-M2.5		X	
DeepSeek R1			X
Mistral Large 3			
Mistral Medium 3			

Table 6: Triadic context split by performance conditions (Low, Medium, High).

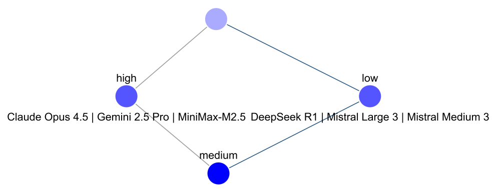
The generated triadic concepts computed from this context are listed below:

- **1.** {Claude Opus 4.5}, {MMLU, SWE-bench}, {H}
- **2.** $\emptyset$, {MMLU, SWE-bench, HumanEval}, {L, M, H}
- **3.** {Mistral Medium 3}, {MMLU, SWE-bench, HumanEval}, {L}
- **4.** {Gemini 2.5 Pro, MiniMax-M2.5}, {MMLU, HumanEval}, {M}
- **5.** {Gemini 2.5 Pro, MiniMax-M2.5, DeepSeek R1, Mistral Large 3}, {MMLU}, {M}
- **6.** {Claude Opus 4.5, Gemini 2.5 Pro, MiniMax-M2.5, DeepSeek R1, Mistral Large 3, Mistral Medium 3}, {MMLU, SWE-bench, HumanEval}, $\emptyset$
- **7.** {DeepSeek R1}, {HumanEval}, {H}
- **8.** {Mistral Large 3, Mistral Medium 3}, {SWE-bench, HumanEval}, {L}
- **9.** {Claude Opus 4.5, Gemini 2.5 Pro, MiniMax-M2.5}, {HumanEval}, {M}
- **10.** {Claude Opus 4.5, Gemini 2.5 Pro, MiniMax-M2.5}, {SWE-bench}, {H}
- **11.** {DeepSeek R1, Mistral Large 3, Mistral Medium 3}, {SWE-bench}, {L}
- **12.** {Claude Opus 4.5, Gemini 2.5 Pro, MiniMax-M2.5, DeepSeek R1, Mistral Large 3, Mistral Medium 3}, $\emptyset$, {L, M, H}

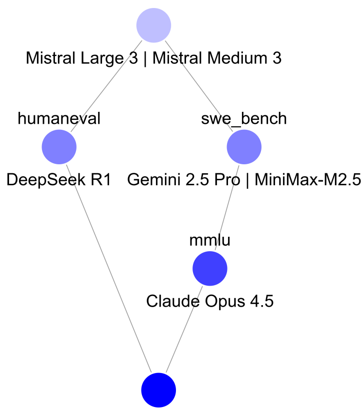
For the local navigation phase, a few key concepts were selected to explore the triadic space. By locking one of the dimensions of a triadic concept, a 2D dyadic context was generated. This allowed for seamless navigation from one node in the concept lattice to another triadic concept.



(a) Locking the Low condition hierarchy.



(b) Navigating to the SWE-bench benchmark scale.



(c) Transitioning to the High performance fork.

Figure 23: Local navigation and paths between the triadic concept states.

As observed in the first layout (Figure 23a), focusing on the Low (L) condition node within the context of Mistral Medium 3 and all three benchmarks shows a clear hierarchical scale. Mistral Medium 3 sits at the bottom because it carries a low value for every single benchmark. Moving up, DeepSeek R1 records a low value specifically for `swe_bench`, while Mistral Large 3 includes the DeepSeek node properties by matching that low software score and adding a low value for `humaneval` as well.

Next, from the DeepSeek node, navigation continues directly into the `swe_bench` attribute concept (Figure 23b), where a new dyadic layout appears. Starting from the root node, the model scores split down the middle into either high or low values with an exact 50% probability (3 high models versus 3 low models). This equal split shows a major discrepancy between the architectural capabilities of these models when facing complex software engineering tasks.

The final state visited is the High (H) condition concept (Figure 23c), which forks cleanly into two distinct paths. On one side, DeepSeek R1 stands out as the top performer at generating Python scripts under `humaneval`. On the other side, Claude Opus 4.5 wins the ultimate challenge by outperforming all five remaining models on the list, sweeping two out of the three main benchmarks with a high accuracy rating.

6 Temporal Concept Analysis

Temporal Concept Analysis (TCA) is used to investigate the temporal evolution of conceptual knowledge by mapping out how formal objects change their states over time. For this task, a geology dataset named "Global Volcano Eruption Intelligence" was selected [Kanchana(2026)]. The dataset contains a comprehensive record of 898 meaningful volcanic eruptions spanning more than 6,300 years, from 4360 BC to the present. The raw data was extracted from the NOAA National Centers for Environmental Information [National Geophysical Data Center and includes over 70 analytical columns.

6.1 Dataset Preparation

To visualize and discuss the results of a Conceptual Time System, the analysis focuses on the timeline and state evolution of a single active volcano over time. After preprocessing the records, the famous Mount Etna—located on the east coast of Sicily, Italy—was found to hold the second-highest number of documented eruptions in the dataset. Following a data-cleaning process where rows with missing key metrics were excluded, the final working sample contains 14 distinct historical eruption timelines for Mount Etna, which are presented in Table 7.

Index	Year	Season	Casualties	VEI Category	Risk Tier	Agent Decoded
1	1169	Winter	True	Unknown	Unknown	Submarine Eruption
4	1329	Summe	False	Severe	Moderate	Inundation / Lahar
5	1536	Spring	True	Severe	Moderate	Tsunami Generated
7	1669	Spring	True	Severe	Moderate	Lava Flow
9	1832	Autumn	False	Explosive	Low	Lava Flow
10	1843	Autumn	True	Explosive	Low	Lava Flow
12	1928	Autumn	True	Gentle	Minimal	Lava Flow
13	1929	Summer	True	Gentle	Minimal	Tsunami Generated
14	1979	Autumn	True	Severe	Moderate	Tsunami Generated
16	1984	Autumn	True	Severe	Moderate	Submarine Eruption
17	1985	Winter	True	Gentle	Minimal	Submarine Eruption
18	1987	Spring	True	Explosive	Low	Tsunami Generated
23	2017	Spring	False	Unknown	Unknown	Phreatic / Phreatomagmatic
24	2018	Winter	False	Explosive	Low	Submarine Eruption

Table 7: Historical eruption timeline sample data for Mount Etna.

To structure this time system into a formal context, four many-valued attributes were selected alongside the continuous time parameters (year, season). The structural meaning, specific ranges, and distinct categorical values of these many-valued attributes are specified in Table 8.

Attribute	Possible Discrete Values (Scale Domain)	Meaning / Operational Source
`has_casualties`	True, False	Boolean indicator for at least one death
`vei_category`	Unknown, Gentle, Explosive, Severe	Volcanic Explosivity Index category labels
`vei_risk_tier`	Unknown, Minimal, Low, Moderate	Risk classification derived directly from the VEI score
`eruption_agent`	I, L, P, S, T	Primary destructive agent code assigned by NOAA

Table 8: Specification and mapping ranges for the many-valued attributes.

The primary physical agents responsible for injuries or environmental impact are decoded using standard NOAA geological classifications [National Geophysical Data Center(2001)] [Gilbert(1997)]:

- **I (Indirect):** Indirect fatalities resulting from post-eruption disease, starvation, exposure, or regional desolation.
- **L (Lava Flow):** Direct property destruction or land coverage caused by molten lava paths.
- **P (Pyroclastic):** High-velocity pyroclastic flows, density surges, and direct volcanic atmospheric blasts.
- **S (Seismic):** Volcanic earthquakes or localized syn-eruptive seismic activity (excluding independent regional tectonic events).
- **T (Tephra):** Volcanic ash, bombs, lapilli, and steam blasts, causing mortality through ballistic impact, asphyxiation, or structural roof collapses.

6.2 Conceptual Time System and Lattice Component Decomposition

To adapt the dataset for Temporal Concept Analysis, the temporal parameters were simplified to focus strictly on structural trends over seasonal periods. The exact days and years were excluded from the formal attributes to prevent computational bloat, while the original calendar months were transformed into their respective meteorological seasons (e.g., February to Winter, March and April to Spring, July and August to Summer, and September, October, and November to Autumn). Within this system, the individual historical eruption instances serve as the fundamental time granules (the units of observation), while the categorical classifications of volcanic impact form the event conditions.

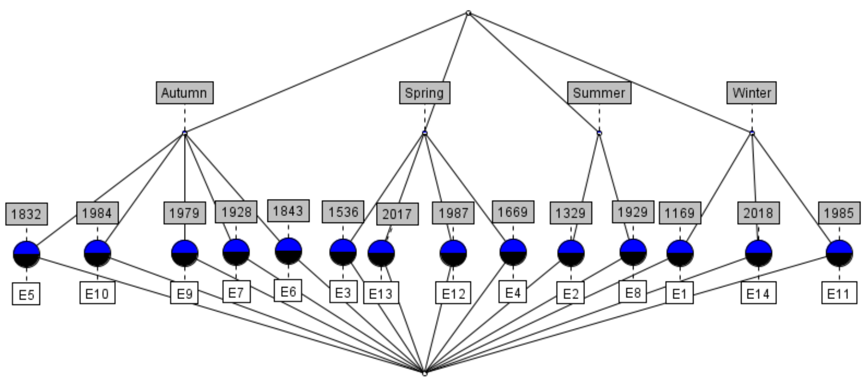


Figure 24: Concept lattice of the temporal component (Seasons).

The first component is the time part context lattice displayed in Figure 24. This structure maps out the flat nominal distribution of the four calendar seasons across our observation units. Since seasons do not overlap naturally for a single instance, the lattice forms a balanced, symmetrical crown where each node handles the isolated seasonal classifications independently before they feed down into the system baseline.

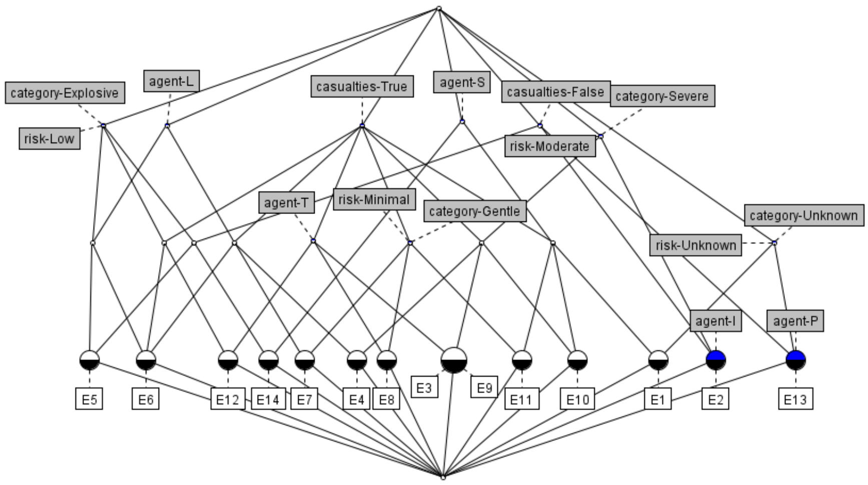


Figure 25: Concept lattice of the event component (Eruption Metrics).

The second component is the event part context lattice shown in Figure 25, which strips away all chronological markers to isolate pure structural implications between the physical characteristics of the eruptions. This lattice highlights how metrics like `vei_category`, `vei_risk_tier`, and the presence of human casualties interconnect, showing which destructive footprints naturally imply or cluster with specific risk levels during Mount Etna's history.

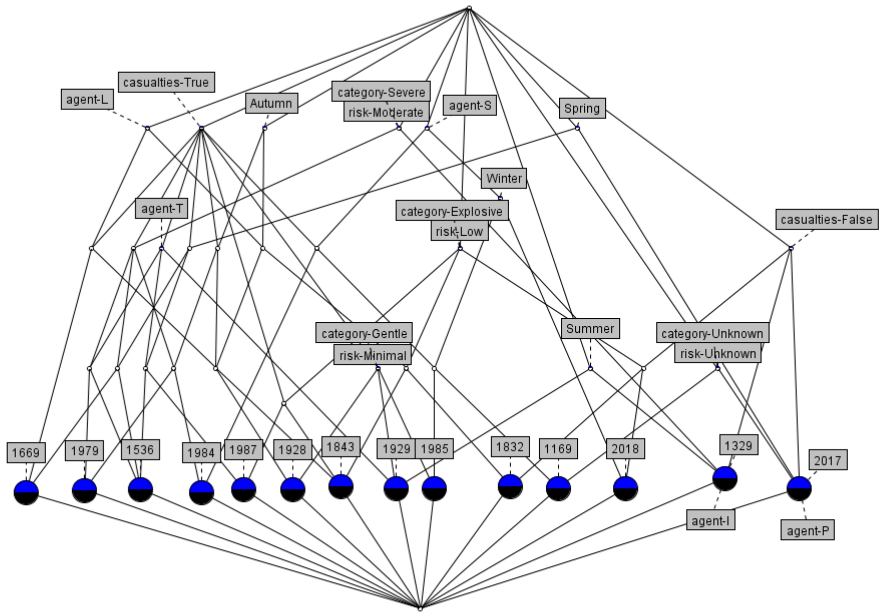


Figure 26: Derived context lattice combining time granules and event conditions.

Finally, by combining both contexts, the complete derived situation space displayed in Figure 26 is gnerated. This comprehensive lattice represents the full intersection of *when* and *what*, mapping out the distinct concept nodes where the individual historical time granules move across the shared event conditions to showcase the complete geological state space of the volcano.

7 Real-Life Application of Formal Concept Analysis: FCART

7.1 Introduction

The Formal Concept Analysis Research Toolbox (FCART) [A.A. Neznanov(2014)] is an integrated software system engineered for information retrieval, data engineering, and knowledge discovery from multi-format data sources. Developed to address the challenges of processing unstructured data, the platform utilizes the mathematical framework of Formal Concept Analysis (FCA) to systematically transform raw datasets into interpretable conceptual structures. FCART serves as an operational research environment where data analysts can define object-attribute-condition relations, execute database queries, and visually navigate complex data systems through formal concept lattices.

7.2 Architecture

The architecture of FCART separates data ingestion, query management, snapshot isolation, and mathematical solving into distinct, modular layers. The software system consists of the following components:

1. **Core Component Layer:**
   - **Multiple-Document User Interface (MDI):** Formulates the primary research environment, integrating a Session Manager to track user operations and an Extensions Manager to control system plugins.

- **Snapshot Profiles Editor (SHPE):** Controls the definition, loading parameters, and structural mapping of isolated data slices.
- **Snapshot Query Editor (SHQE):** Facilitates the execution of analytic selection queries against the intermediate data layer.
- **Query Rules Database (RDB):** Stores the logical rules, dependencies, and parameters governing context-scaling operations.
- **Session Database (SDB):** Records the exact chronological steps, computational artifacts, and analytical states within a research session.
- **Report Builder:** Compiles calculated concept lattices, association rules, and statistical data directly into publication-ready templates.

2. **Local Data Storage (LDS):** Acts as an intermediate repository where extracted data tables are stored, structured, and prepared for analytical processing.

3. **Internal Solvers and Visualizers:** Dedicated programmatic modules that execute scaling transformations, calculate formal concepts, and render interactive graphical lattices.

4. **Plugins, Scripts, and Templates:** Programmable add-ons that extend the core toolbox functionality to handle specialized text mining and external data processing scripts.

7.3 Main FCA Workflow

From an operational perspective, the analytical process within FCART is divided into four distinct stages. The user can import, export, or append artifacts to the final report at any stage of the pipeline:

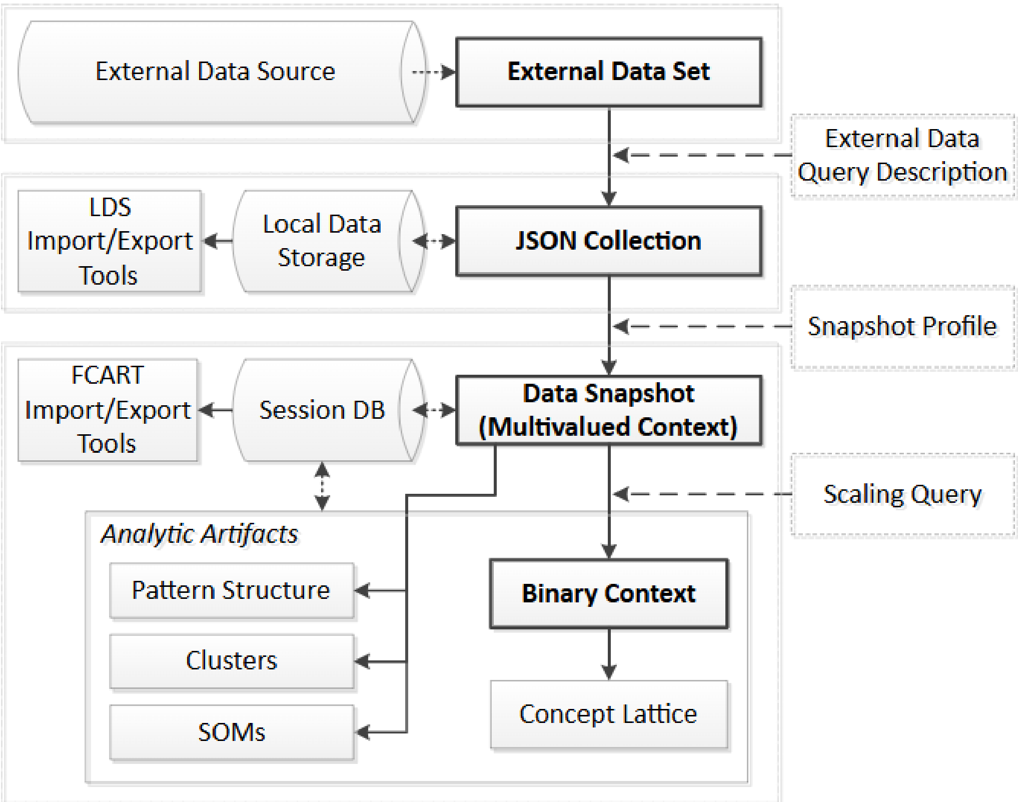


Figure 27: Main FCA Workflo in FCART [Neznanov et al.(2014)Neznanov, Ilvovsky and Parinov]

1. **Stage 1 — Filling the Local Data Storage (LDS):** Data tables are extracted from external relational databases (SQL), XML files, or JSON repositories. This process is governed by an External Data Query Description (EDQD), which defines the communication parameters with external data browsers.

2. **Stage 2 — Loading Data Snapshots:** A specific subset of preprocessed data is moved from the LDS into the active session. This snapshot is defined by a Snapshot Profile and forms a structured spreadsheet table where textual and quantitative attributes are annotated.

3. **Stage 3 — Conceptual Scaling:** The many-valued attributes in the snapshot are transformed into a binary formal context. This transformation is executed according to criteria defined in a specialized Scaling Query.

4. **Stage 4 — Lattice Building and Visualization:** The internal mathematical solvers process the binary context. The system generates the formal concept lattice and extracts analytical parameters, allowing the user to visually navigate the resulting concept hierarchy.

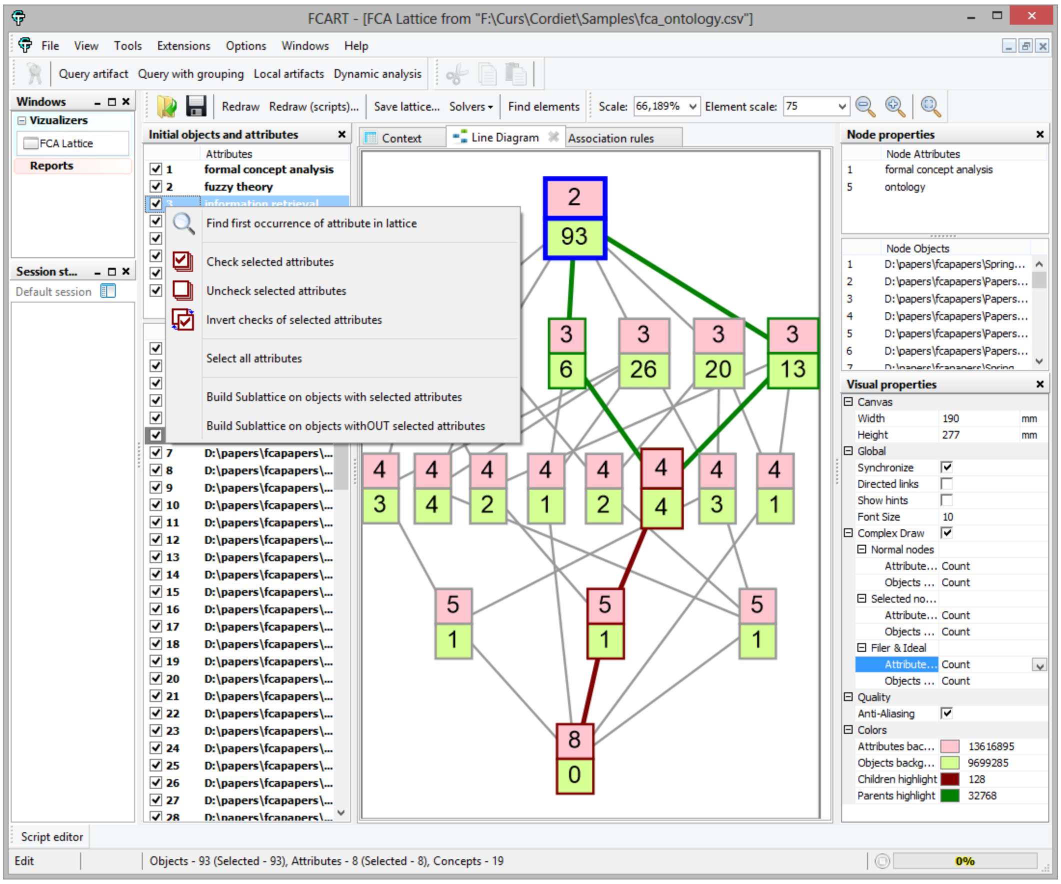


Figure 28: FCART Search

7.4 Comparison with Existing Systems

To evaluate the functionality and performance of FCART, its capabilities must be compared directly against traditional analytical suites and specialized Formal Concept Analysis (FCA) tools. Large-scale industrial applications like IBM i2 Analyst's Notebook or QSR NVivo do not provide native FCA utilities and operate on entirely different structural data analysis methodologies. Therefore, assessing FCART requires using specific architectural criteria: basic functionality (context editing, lattice drawing, sublattice construction, and association rules generation), performance under high operational loads, preprocessing tools, format versatility, visualization control, reporting, and ease of system extensibility.

Most well-known FCA software tools feature highly distinct, isolated strengths. Concept Explorer (ConExp) served as an important milestone due to its robust default lattice rendering and unique visualization modes. Galicia introduced the multi-context MultiFCA approach, while ToscanaJ integrated nested lattices with a schema editor designed for relational databases. FcaStone was built primarily for low-level file conversions. Despite these unique features, a widespread issue in the FCA software landscape is legacy stagnation; development for the majority of end-user toolkits ceased years ago, with Coron's last release occurring in 2010 and other major projects completely halting official updates after 2006. Today, only ToscanaJ and Conexp-clj represent actively maintained external projects.

This lack of modernization creates severe performance limits when interactively managing large datasets. Cross-platform GUIs and heavy reliance on high-level languages like Java cause conventional programs to lag or crash when manipulating lattices containing more than 8,000 concepts. This scalability bottleneck becomes clear when testing a real-world context of 707 objects and 257 attributes, which builds a complex lattice of 10,568 concepts. Under these conditions, ConExp successfully generates the concepts but consumes 90.0 MB of memory and completely fails to render the visual layout of the lattice. ToscanaJ's Siena module manages to calculate the concepts and produce a layout, but it allocates a massive 203.2 MB of memory and suffers from severe lag, running very slowly even when simply displaying the flat initial context.

In contrast, FCART processes the identical 10,568-concept layout smoothly, consuming only 23.5 MB of memory while maintaining fluid, normal interactive manipulation of both the context grids and the concept nodes. The current architecture of FCART can comfortably generate, visualize, and manipulate large lattices exceeding 16,000 concepts in real-time active sessions. By running on top of a modern software platform, FCART eliminates the GUI overhead of older cross-platform systems, providing analysts with powerful integrated preprocessing tools, comprehensive automated reporting, and two distinct levels of programmatic extensibility.

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